

Storytelling with data

Chapter resources

Practice quiz

i Materials for this lecture

The slides for this lecture reproduce figures from Cole Nussbaumer Knaflic's *Storytelling with Data*, which cannot be redistributed, so they are **shared directly by the teacher** rather than published here. The book itself is available in the department library and is the reference for the whole lecture.

An analysis that nobody understands has not been completed. This lecture is about the last step: turning a set of findings into something an audience can follow, act on, and challenge.

It sits deliberately at the end of Block 1, right after the [causal chapter](#). By this point you can establish what the data supports; the risk from here on is that a confident chart says more than the analysis does. Most misleading data graphics are not produced by liars — they are produced by people who wanted to make a point clearly.

What the lecture covers

The material follows the structure of Knaflic's book:

1. **Understand the context.** Who is the audience, what do they already know, what decision are they going to make, and what exactly do you need them to do? A chart for yourself and a chart for a stakeholder are different objects.
2. **Choose an effective visual.** Match the display to the question — the comparison you want the reader to make should be the easiest one to make on the page. Chapter 2 of these notes covers the mechanics; this is about the choice.
3. **Eliminate clutter.** Every element that is not carrying information is competing with the ones that are: gridlines, borders, legends that could be direct labels, decoration, the third dimension of a bar chart.
4. **Focus attention.** Colour, size and position are the tools that say *look here*. They work only if they are used sparingly: if everything is highlighted, nothing is.
5. **Think like a designer.** Affordance, accessibility and aesthetics — make the chart look like what it does, make it readable to a colour-blind reader and on a projector, and accept that a tidy chart is trusted more than an untidy one.
6. **Tell a story.** Narrative structure — situation, complication, resolution — applied to an analysis, so the audience remembers the point rather than the plot.

The part that belongs to this course

Two obligations run alongside the craft, and they are what we will assess.

A chart must not claim more than the analysis supports. The most common failure in project presentations is a slide showing a clean association with a title written in causal language. If chapter 11 could not close the backdoor paths, the title cannot either. This is why the project's slide deck must include a "What could confound this?" slide: it forces the question onto the same page as the claim.

Uncertainty is part of the result, not a caveat to be dropped for clarity. A point estimate shown without its interval is a cleaner chart and a worse one. Error bars, a confidence band, or a sentence stating the interval — one of the three has to be there.

! For the project

Your presentation is the oral part of the exam. It is assessed on whether the audience can follow the argument, whether the charts are readable, whether the limitations are stated, and whether each member can answer questions about the work. Fifteen slides at most. See the [project page](#).

References

- Knaflic, Cole Nussbaumer. *Storytelling with Data: A Data Visualization Guide for Business Professionals*. Wiley, 2015. **Available in the department library.**
- Tufte, Edward. *The Visual Display of Quantitative Information*. 2nd ed., 2001.
- Cairo, Alberto. *How Charts Lie*. Norton, 2019.
- Wilke, Claus O. *Fundamentals of Data Visualization*. O'Reilly, 2019. Free online: <https://clauswilke.com/dataviz/>